asphalt pavement, this problem can be economically met by first building the thickness required to cope with initial and short-term loading and then adding, when needed, additional layers of asphalt to increase total pavement thickness. This procedure is called stage construction. It avoids extensive investment in the beginning; when a new asphalt pavement layer is added, the wearing surface is equal to or better than the original.

When road-building funds are limited, a strong all-weather asphalt base can be placed and opened to traffic for a year or more. When additional construction funds are available, the asphalt surface course can be added. Similarly, in new housing developments, asphalt base can be placed on all new streets to serve until the development is finished and heavy construction traffic ceases. The asphalt surface is then added after first repairing any damage or settlement suffered by the base while supporting the construction traffic. The streets are then at full strength and in an attractive condition appropriate to a new living area.

Many examples of planned stage construction are evident in the interstate highway system. Asphalt pavements have been built with sufficient thickness to serve for the first few years, with funds programmed to add strengthening thickness when required by increases in traffic.

1.5 Overview of the Handbook

The handbook is divided into the following five sections:

- Chapters 2, 3, and 4 cover the materials used to build HMA pavements and how to design those materials.
- Chapters 5 and 6 cover the design of an HMA pavement, starting with the subgrade or foundation soils and then the actual thickness design of the HMA layer.
- Chapters 7, 8, 9, and 10 cover the construction of the asphalt layer, starting with production and moving through to quality control of the HMA.
- Chapters 11, 12, 13, and 14 present information on the management of the HMA pavement, including maintenance and rehabilitation.
- Chapter 15 covers special uses of asphalt and HMA.

Chapter 2—Petroleum Asphalt

This chapter discusses the manufacture of the asphalt binder to obtain desired properties. It also discusses the historical tests that have been used, and are still used in many parts of the world, to describe the properties of asphalt cement. It describes in detail the Performance Graded (PG) system for specifying asphalt binders and the testing associated with that system. It also describes the manufacture and testing of emulsified and cutback asphalts. A brief discussion of air blown asphalts is provided. Finally, the steps that must be taken to ensure that those working around asphalt cements do it in a safe manner are also provided.

Chapter 3—Mineral Aggregates and Mixture Additives

This chapter discusses aggregate processing and aggregate properties. Included in this discussion are sources of aggregates and the properties of those aggregates as they affect the quality and performance of an HMA mixture. The tests used to evaluate the quality and properties of various aggregate sources are also discussed. The tools needed to analyze a composite aggregate blend are presented, and example problems are provided. The properties and handling of recycled asphalt pavement (RAP) materials are discussed, along with the proper procedures for sampling and testing RAP. Recent developments in the construction of HMA pavements have introduced the use of mixture additives. The use of these materials is discussed.

Chapter 4—Hot Mix Asphalt Mix Design

This chapter presents the basic concepts of HMA mix design and the steps to calculate the volumetric properties of an HMA mixture. It covers the Superpave mix design procedure in detail and introduces the background of the Marshall and Hveem mix design procedures. This chapter also covers the use of reclaimed asphalt pavement (RAP) in an HMA mix

design and the design of specialized HMA mixtures such as open-graded friction courses (OGFC), stone matrix asphalt (SMA), and permeable base layers. The use of performance testing for the evaluation of HMA materials is briefly introduced.

Chapter 5—Soils and Drainage for Asphalt Pavements

This chapter presents a background on the characteristics of soils as they relate to the design and construction of HMA pavements. The test procedures used to evaluate the strength of soils are reviewed, and the development of a proper drainage system for an HMA pavement is covered.

Chapter 6—Asphalt Pavement Structural Design

This chapter presents an overview of the various Asphalt Institute pavement thickness design methods and how those methods relate to other thickness design methods used for the design of HMA pavements. This description includes a discussion about the required inputs, such as environmental (temperature, rainfall), traffic (type, mix of traffic, and traffic volume), and subgrade strengths. It also discusses how to handle very heavy wheel loads in the development of a pavement design for a specialized use such as container yards or airfields.

Chapter 7—Manufacture of Hot Mix Asphalt

This chapter presents a discussion of the HMA manufacturing process. It covers the components of an HMA manufacturing facility, how aggregate and RAP are fed in a controlled manner into the HMA plant, how it is dried, and how the dust generated by the production process is controlled. It details the operation of both a batch plant and a drum-mix plant, and it provides a detailed discussion of troubleshooting problems that might develop during the production of the HMA.

Chapter 8—Hot Mix Asphalt Paving

This chapter presents the process of actually constructing an HMA pavement including: subgrade and base course preparation for a new pavement; placement of the HMA layers; and pavement surface preparation for an HMA overlay over an existing pavement. The operation of the paving machine is covered in detail, along with steps that can be taken to prevent problems during the placement of HMA mixtures.

Chapter 9—Hot Mix Asphalt Compaction

This chapter covers the concepts behind the proper compaction of an HMA pavement. The principles of compaction that are discussed include the following: the effect of HMA properties; environmental effects; layer thickness; subgrade and subbase as it affects compaction of the HMA; and the time available for compaction. A discussion of the different types of compaction equipment is provided, along with how that equipment should be used to properly compact an HMA pavement.

Chapter 10—Quality Control and Acceptance of Hot Mix Asphalt

This chapter presents a discussion of quality-control practices necessary to ensure that the contractor produces a high-quality HMA mixture. It covers sampling and testing both at the HMA production facility and during the placement process. Included is a discussion about the process

used to analyze the data being developed as a part of the quality control and acceptance process. It also discusses the use of performance and warranty specifications for construction of HMA pavements.

Chapter 11—Pavement Management

This chapter discusses how the pavement management process works to ensure an adequate pavement performance over time. The components of a pavement management system and how that system can be used by an agency as a tool in the development of a pavement preservation system are presented. The steps for conducting a pavement evaluation of an existing pavement and the procedures to accomplish a pavement distress survey are covered.

Chapter 12—Maintenance of Asphalt Pavements

This chapter presents a detailed discussion of the materials and procedures that can be used to properly maintain an HMA pavement. It includes a discussion about what is needed to maintain a pavement and why various procedures are used. The chapter covers patching, cracking sealing, and asphalt surface treatments (fog seals, slurry seals, micro-surface, and chip seals).

Chapter 13—Rehabilitation Methods for Asphalt Pavements

This chapter discusses the proper methods for rehabilitating an HMA pavement. It contains a detailed discussion of the steps that should be taken before placing an overlay including patching, interlayers milling and inlay procedures, hot-in-place recycling, cold-mix recycling, and full-depth reclamation.

Chapter 14—Rehabilitation of Portland Cement Concrete Pavements with HMA

This chapter presents the steps that can be taken to rehabilitate a failed portland cement concrete pavement. A review of the process of reflection cracking is provided. The use of saw-cut and seal, crack/break and seat, and rubbilization are discussed in detail.

Chapter 15—Special Uses of Asphalt and HMA

This chapter is a brief discussion about the ways asphalt is used for other than highways. For example, the HMA is used for hydraulic structures, recreational facilities, leaching pads, and racetracks.

CHAPTER 2

PETROLEUM ASPHALT

This chapter discusses liquid asphalt binder (also called bitumen) as a material. It is derived from petroleum processing. Properties, characteristics, and commonly used tests and specifications are discussed. In the interest of brevity, all the test details and procedures are not given here but can be found in other sources.

Section 2.1 Refining and Properties of Asphalt Binders
- 2.1.1 Introduction
- 2.1.2 Refining Crude Petroleum
- 2.1.3 Characteristics of Asphalt Binders

Section 2.2 Historical Perspective: Asphalt Cement
- 2.2.1 Introduction
- 2.2.2 Tests for Asphalt Cements
- 2.2.3 Specifications for Asphalt Cements

Section 2.3 Dominant Practice: Performance Graded (PG) Asphalt Binder
- 2.3.1 Introduction
- 2.3.2 Asphalt Behavior
- 2.3.3 PG Asphalt Binder Tests
- 2.3.4 PG Asphalt Binder Specification

Section 2.4 Emulsified and Cutback Asphalts
- 2.4.1 Introduction
- 2.4.2 Tests for Emulsified Asphalts
- 2.4.3 Specifications for Emulsified Asphalts
- 2.4.4 Tests for Cutback Asphalts
- 2.4.5 Specifications for Cutback Asphalts

Section 2.5 Air-Blown Asphalts
- 2.5.1 Introduction
- 2.5.2 Properties, Specifications, and Tests

Section 2.6 Asphalt Modification and Additives

2.6.1 Reasons for Using Additives and Modification
2.6.2 Types of Modifiers and Additives
2.6.3 Additional Tests for Modified Asphalt Binders

Section 2.7 Supplementary Asphalt Topics

2.7.1 Control of Asphalt Temperatures
2.7.2 Asphalt Sampling
2.7.3 Temperature-Volume Relationship
2.7.4 Frequently Asked Asphalt Questions

Section 2.8 Safety Procedures for Hot Asphalt Binder

Note: Although the standard practice in this manual is to list temperatures in English units first, followed by metric units in parenthesis, the PG System uses metric units exclusively.

2.1 Refining and Properties of Asphalt Binders

2.1.1 Introduction

According to the American Society for Testing and Materials (ASTM), asphalt is a dark brown to black cementitious material in which the predominating constituents are bitumens, which occur in nature or are obtained in petroleum processing.

Asphalt is present in varying proportions in most crude petroleum sources. Although asphalts also occur as natural asphalts, rock asphalts, and lake asphalts, the majority of asphalt material used today is obtained from petroleum through various refining processes. As such, this chapter provides information on petroleum asphalt binders.

In Europe, the terms *bitumen* and *asphaltic bitumen* are used and are synonymous with the term *asphalt* used in North America. Outside North America, the term *asphalt* is used to describe mixtures of bitumen with mineral aggregates. To avoid confusion, in North America liquid asphalt is often also referred to as "asphalt cement" or, more generically, as "asphalt binder."

Asphalts are viscous liquids or solids essentially consisting of hydrocarbons and their derivatives, which are soluble in carbon disulfide. They are substantially nonvolatile at ambient temperatures and soften gradually when heated. Asphalts are often referred to as viscoelastic materials, behaving as elastic solids at low temperatures and viscous liquids at high temperatures.

Asphalts should not be confused with coal-tar products such as coal tar or coal-tar pitches. The latter are manufactured by the high temperature carbonization of bituminous coals and differ from asphalts substantially in composition and physical characteristics. The differences between asphalts and coal-tar products are well defined in industry literature.

Similarly, asphalts should not be confused with petroleum pitches, which are often highly aromatic residues, produced by thermal cracking, coking, or oxidation from selected petroleum fractions. Petroleum pitches are primarily used as binders in the manufacture of metallurgical electrodes.

2.1.2 Refining Crude Petroleum

Most asphalt produced in the United States is the result of the crude oil distillation process. Even though vacuum distillation is the basic unit of operation, additional manufacturing methods can also be used. The choice of manufacturing methods depends on the properties of crude available. Because crude oil chemical composition varies significantly, the properties and yield of residuum can also vary over a wide range.

As API (American Petrochemical Institute) gravity of crude oils increases, the yield of residuum decreases and the yield of distillate subsequently increases. The industry defines crude oil as "heavy" when its API gravity is less than about 25. These low API gravity crude oils are also called "sour" crude oils if their sulfur content is high. If a crude oil has an API gravity greater than about 25 and is generally low in sulfur content, it is called a "light" and "sweet" crude oil. Most fully integrated refineries are designed to maximize distillate products and are not generally adapted to process heavy crude oils, possibly because of the investment required in processes to extract the sulfur. This limitation can restrict the refiners' options for crude selection and ability to produce asphalts. On the other hand, asphalt refineries are designed to maximize the production of heavy petroleum products and the refiner intentionally selects heavy crude oils for processing.

Several manufacturing methods are available to produce specification asphalts depending on the crude source(s) and processing capabilities available. Often, a combination of processes is selected. Table 2.1 provides a summary of the manufacturing methods that may be used by an asphalt producer.

Table 2.1
Methods Used to Produce and Process Asphalt Materials

Production/Process	Base Materials	Product
Atmospheric and Vacuum Distillation	Asphalt-based crude or crude mix	Asphalt cements
Blending	Hard and soft asphalt	Asphalt cements of intermediate consistency
	Asphalt cements and petroleum distillates	Cutback asphalts
Air-Blowing	Asphalt flux	Asphalt cements, roofing asphalts, pipe coatings, special membranes
Solvent Deasphalting	Vacuum residuum	Hard asphalt
Solvent Extraction	Vacuum residuum	Asphalt components (asphaltenes, resins, oils)
Emulsification	Asphalt, emulsifying agents, and water	Emulsified asphalts
Modification	Asphalt and modifiers (polymers, chemicals, etc.)	Modified asphalts

The most common practice for producing asphalt cements is straight reduction to grade from crude oil or a crude blend using atmospheric and vacuum distillation. The petroleum asphalt flow chart (see Figure 2.1) shows, in general, the flow of crude petroleum through a refinery. The emphasis is on that part of the process related to the refining and production of asphalt cement.

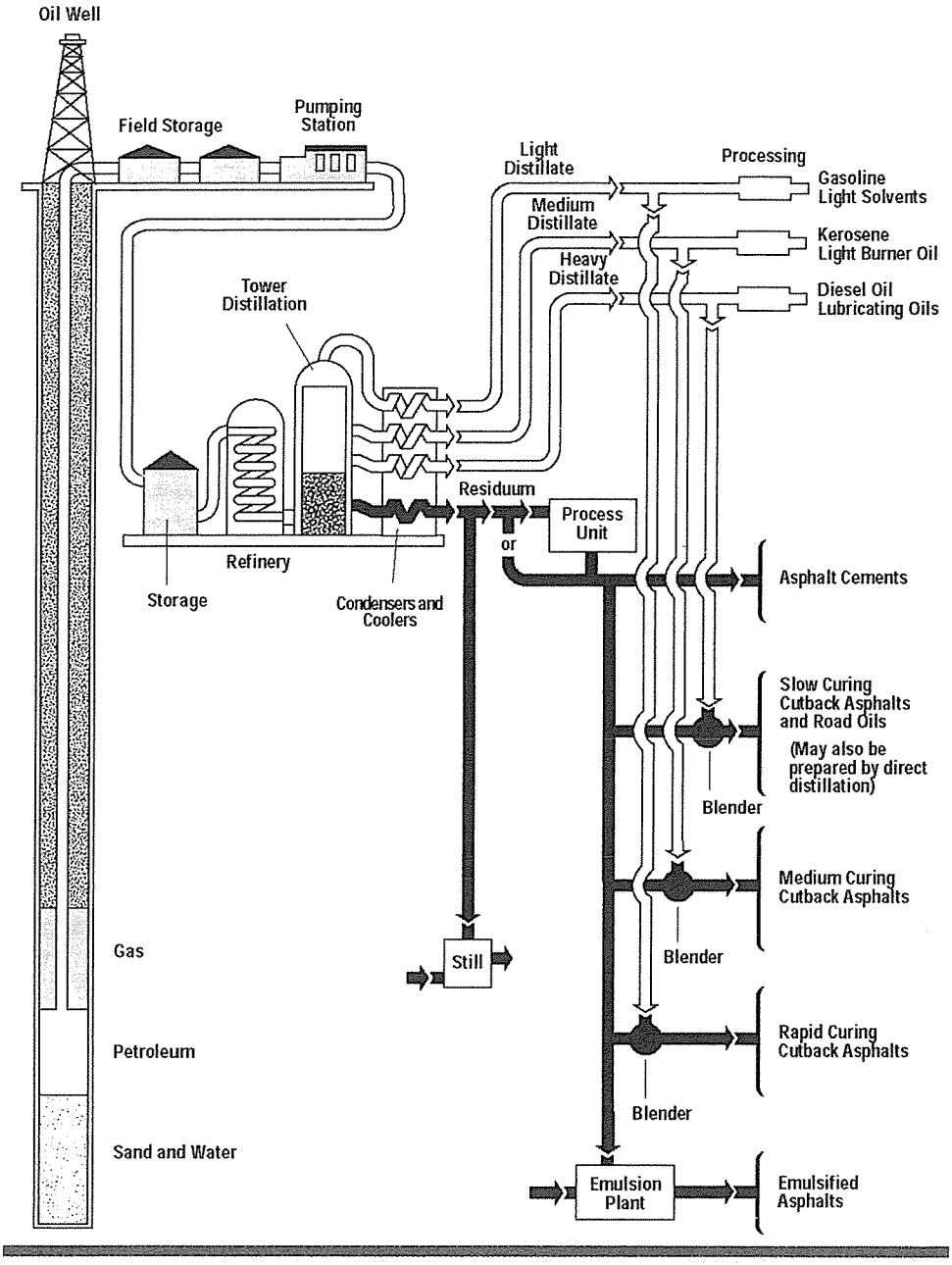


Figure 2.1 Petroleum Asphalt Flow Chart

Atmospheric distillation is used to separate various "overhead" crude fractions and a non-boiling component known as an atmospheric residue. Crude flows continuously through a furnace that heats the oil and strips out any light materials present. The heated crude flows through an atmospheric tower where light components flash to the upper reaches of the tower and subsequent lower boiling components reach lower sections of the tower. Different boiling components are collected as they condense

and are removed for further processing. The overhead fractions include gasoline (60–325°F/15–165°C), naphtha (300–400°F/150–200°C), kerosene (325–500°F/165–260°C), light gas oil (450–600°F/230–315°C), and heavy gas oil (575–700°F/300–370°C).

The non-boiling atmospheric residue is introduced into a vacuum distillation unit in which several high boiling distillate fractions are produced, leaving a vacuum residue. The overhead fractions are gas oils (650–750°F/345–400°C and 700–850°F/370–455°C). The vacuum residue is used as the base material for the production of asphalt. The atmospheric equivalent vapor temperature to yield the vacuum residue is about 1000°F (540°C).

The vacuum residue is also the feed stock of several operations that are designed to increase the yield of refined products. Some of these processes yield hard asphalts that can be used as blends with other components, such as atmospheric and vacuum residua, to produce asphalts.

The processes that produce hard asphalt components for blending include solvent extraction (ROSE), solvent deasphalting (SDA) or propane deasphalting (PDA), and air blowing or blowing rectification (OX). Practically, refineries that produce asphalt cements may use all of these processes in combination and will produce several grades of asphalts, which are further blended to produce intermediate grades.

Air-blown asphalts are produced by blowing air at high temperatures through asphalt of the proper consistency. These asphalts are discussed in Section 2.5, "Air-Blown Asphalts."

2.1.3 Characteristics of Asphalt Binders

For engineering and construction purposes, three properties or characteristics of asphalt are important: consistency (also often called viscosity), purity, and safety.

Consistency

Asphalts are thermoplastic materials, meaning that they liquefy when heated and solidify when cooled. They are characterized by their consistency or ability to flow at different temperatures. *Consistency* is the term used to describe the viscosity or degree of fluidity of asphalt at any particular temperature. The consistency of an asphalt binder varies with temperature; therefore, it is necessary to either define an equivalent temperature or an equivalent consistency when comparing the temperature-consistency characteristics of one asphalt binder with another. In some specifications, asphalt binders are graded based on their consistency at a standard temperature; in other specifications, asphalt binders are graded based on the temperature at which they meet a standard consistency.

When asphalt is exposed to air in thin films at elevated temperature—which occurs during the mixing process with aggregate in a hot mix asphalt mixing facility—the asphalt hardens. This means that the consistency or viscosity of the asphalt increases. While this increase in stiffness is

expected and allowable to a certain extent, a great deal of damage through hardening can be caused to the asphalt binder by carelessly overheating the asphalt during mixing. Excessive hardening may cause the asphalt pavement to experience early cracking failures, significantly shortening the life of the finished asphalt pavement.

Common methods for determining the consistency of an asphalt binder and its importance in standard specifications will be discussed in following paragraphs.

Purity

Asphalt binders are composed almost entirely of bitumen, which by definition is entirely soluble in carbon disulfide. Refined petroleum asphalts are almost pure bitumen and are usually more than 99.5 percent soluble in carbon disulfide. Organic material and impurities, if present, are inert.

Safety

Asphalt foaming is a safety hazard, and specifications usually require that asphalt not foam at temperatures up to 347°F (175°C). Normally, asphalt is free of water or moisture as it leaves the refinery. However, transports loading asphalt may have some moisture present in their tanks. If any water is inadvertently present, that water will become steam when the hot asphalt is added to the tank; the steam will cause the asphalt to foam.

Asphalt binder, if heated to a high enough temperature, releases fumes that flash in the presence of a spark or open flame. The temperature at which this occurs is called the flash point and is well above the temperatures normally used in paving operations. However, to be certain of an adequate margin of safety, the flash point of the asphalt is measured and controlled.

2.2 Historical Perspective: Asphalt Cement

2.2.1 Introduction

To verify that an asphalt cement had the desired characteristics (consistency, purity, and safety), it was necessary to develop tests and specifications that related to these characteristics. Some of the tests described in the next few sections have existed since petroleum asphalt began to be widely used in the early 1900s.

2.2.2 Tests for Asphalt Cements

Viscosity

The absolute viscosity test at 140°F (60°C) employs a capillary tube viscometer. Two types are most commonly used: the Asphalt Institute vacuum viscometer (see Figure 2.2) and the Cannon-Manning vacuum viscometer. Each viscometer is calibrated by using standard calibrating oils. A set of calibration constants are developed for each viscometer and used as described later in this chapter. Viscometers are usually calibrated by the manufacturer, and calibration constants are furnished with each tube.

The viscometer is mounted in a thermostatically controlled, constant temperature bath (see Figure 2.3). Water is often used as the liquid medium in the bath because the test temperature is below its boiling point. Preheated asphalt is then poured into the large side of the viscometer until its level reaches the filling line. The filled viscometer tube is then placed back in an oven operating at the same temperature used to heat the asphalt binder for a short time. Finally, the viscometer is placed in the constant temperature bath for a prescribed time to allow the system to reach an equilibrium temperature of 140°F (60°C).

At 140°F (60°C), a paving-grade asphalt binder is too viscous to flow readily through a capillary tube viscometer. Therefore, absolute viscosity test equipment uses a vacuum pump to apply a consistent partial vacuum to the small side of the viscometer to induce flow.

After the bath, viscometer, and asphalt binder have stabilized at 140°F (60°C), the prescribed vacuum is applied and the time in seconds required for the asphalt binder to flow between two timing marks (delineating a section of capillary tube also called a "bulb") is precisely measured. Multiplying the measured time to flow through this bulb by the calibration constant for the particular bulb of the viscometer gives the value for viscosity in *poises*, the standard unit for measurement of viscosity.

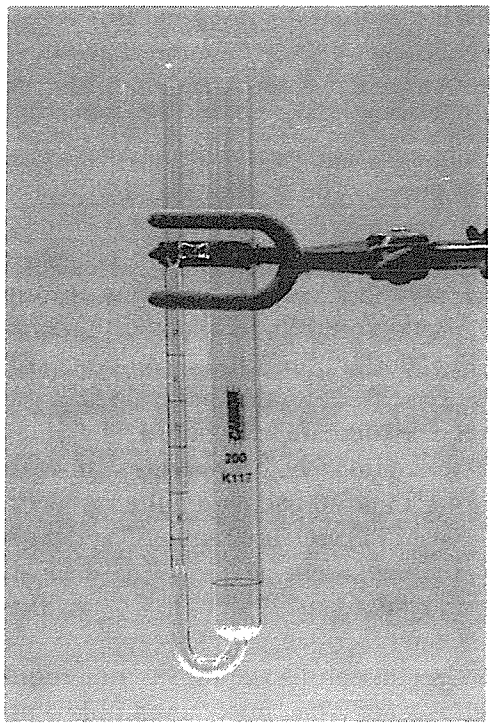


Figure 2.2 Asphalt Institute Vacuum Viscometer Tube

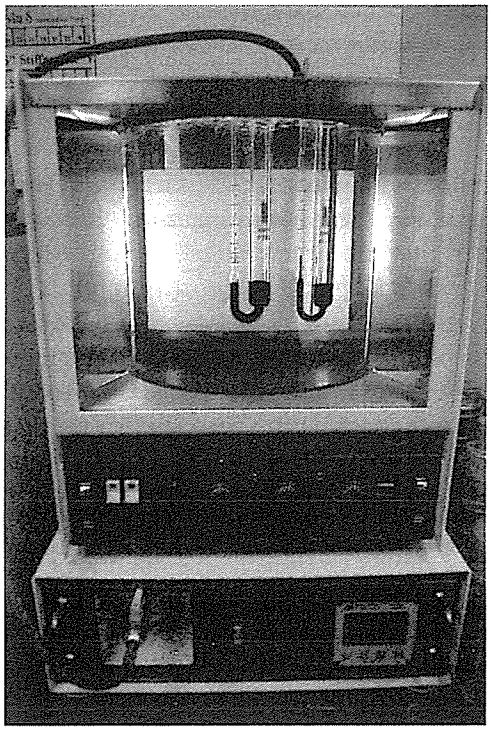


Figure 2.3 Absolute Viscosity Bath with Tubes

There are several different sizes of Asphalt Institute vacuum viscometers, with different size capillary tubes. Common sizes used for paving-grade asphalt binders are identified as 50, 100, and 200 tubes. Higher number tubes are used for asphalt binders with greater viscosity. There are also 400 and 800 tubes, usually followed with an *R* designation, that are used for roofing asphalts (hence the *R* designation) and very viscous paving-grade asphalts. Each viscometer also has several timing marks, delineating several bulbs. To ensure the repeatability of viscosity measurements, the test procedure indicates that the time to flow through a bulb (between successive timing marks) should be at least 60 seconds before calculating viscosity (multiplying time by the calibration constant for the appropriate bulb). If the time to flow through the last bulb in a viscometer is less than 60 seconds, then the asphalt binder should be retested using the next smaller numbered tube. Similarly, if the time to flow through the first bulb in a viscometer is more than 60 seconds, then good practice dictates that the asphalt binder should be retested using the next higher-number tube.

Unlike viscosity measurements at 140°F (60°C), paving-grade asphalt binders are sufficiently fluid at 275°F (135°C) to flow through capillary tubes under gravitational forces alone. This type of viscosity is called kinematic viscosity. In this testing, a vacuum is not required, but a different type of viscometer is used. The most commonly used is the Zeitfuchs cross-arm viscometer (see Figure 2.4).

As with absolute viscosity, each viscometer is also calibrated using standard calibrating oils. A set of calibration constants are developed for each viscometer and used as previously described. Normally, viscometers are calibrated by the manufacturer and calibration constants are furnished with each tube.

Because kinematic viscosity tests are conducted above the boiling point of water at 275°F (135°C), a suitable clear oil must be used as the bath medium. The viscometer is mounted in the bath (see Figure 2.5), and asphalt is poured into the large opening until it reaches the filling line. The system is then allowed to reach equilibrium temperature as before. A slight pressure is applied to the large opening, or a slight vacuum is applied to the small opening, to start the asphalt flowing over the siphon section just above the filling line. The asphalt will then flow downward in the vertical section of capillary tubing by gravitational forces. A timer is started when the asphalt reaches the first timing mark and stopped when it reaches the second. The timer interval, multiplied by a calibration factor for the viscometer, gives the kinematic viscosity in centistokes.

There are several different sizes of the cross-arm viscometers, with different size capillary tubes. Common sizes used for paving-grade asphalt binders are identified as 5, 6, and 7 tubes. Higher number tubes are used for asphalt binders with greater viscosity. As with the absolute viscosity test, in order to ensure the repeatability of viscosity measurements, the time to flow between the timing marks should be at least 60 seconds. If the time to flow is less than 60 seconds, then the asphalt binder should be retested using the next smaller number tube.

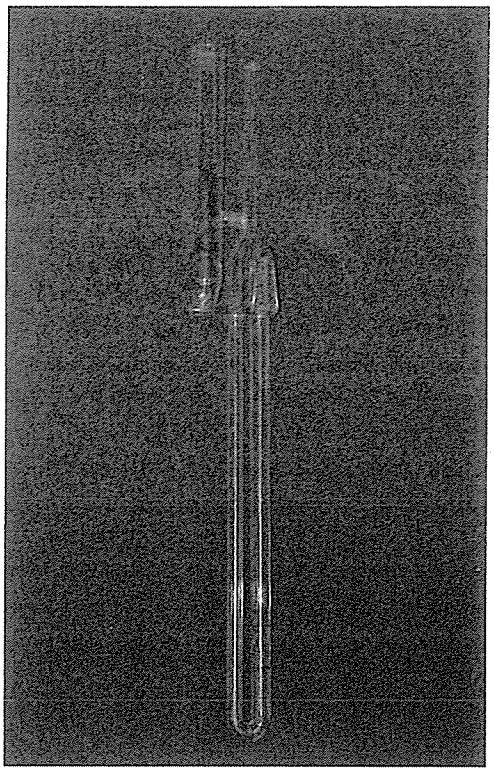


Figure 2.4 Cross-arm Viscometer Tube

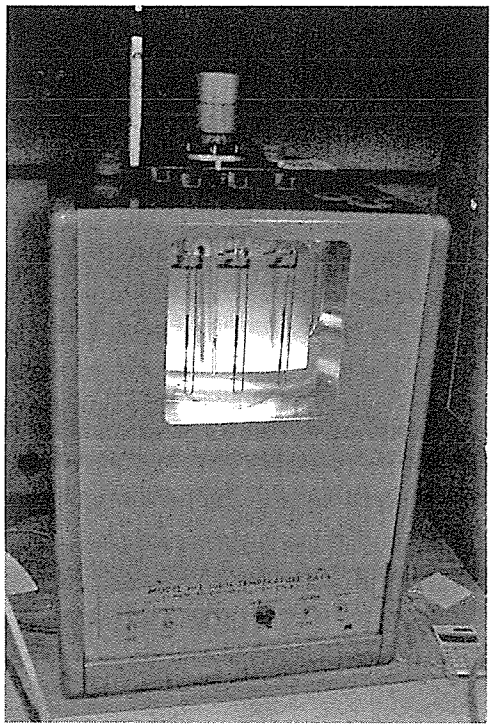


Figure 2.5 Kinematic Viscosity Bath with Tubes

As an alternative, the rotational viscosity test may also be used to determine the viscosity of an asphalt binder at 275°F (135°C). The rotational viscosity equipment and test procedure are described in Section 2.3, "Dominant Practice: Performance-Graded (PG) Asphalt Binder."

Note that viscosity at 275°F (135°C) is expressed in centistokes while viscosity at 140°F (60°C) is expressed in poises. Gravity induces the flow in the kinematic viscosity test (results in centistokes), and the density of the asphalt binder affects its rate of flow through the capillary tube. In the absolute viscosity test (results in poises), flow through the capillary tube is induced by a partial vacuum where gravitational effects are negligible. The units of the two tests, poises and stokes or centipoises and centistokes, are related to each other by the density of the asphalt binder.

Viscosity measurements made at the two test temperatures are sometimes plotted on a viscosity-temperature graph with a line connecting the two points. The slope of this line is an indication of the temperature susceptibility of the asphalt binder—the greater the slope of the line, the greater the temperature susceptibility of the asphalt binder. Asphalt binders that have high temperature susceptibility may be soft at hot temperatures and hard at cold temperatures—a poor combination for successful performance at in-service pavement temperatures.

Additionally, the temperature-viscosity relationship of an asphalt binder can be important in determining appropriate temperature ranges for laboratory mixing and compaction of asphalt mixtures. Laboratory mixing and compaction temperatures of HMA are discussed in Chapter 4, "Hot Mix Asphalt (HMA) Design."

Penetration

The penetration test is one of the oldest, empirical tests for measuring the consistency of an asphalt binder. Figure 2.6 shows the standard penetration test equipment.

To conduct the penetration test, an asphalt binder sample is heated to an appropriate pouring temperature and poured into a test container—usually a 3-ounce (88-milliliter) tin (for standard paving-grade asphalt binders). After a specified conditioning period, the asphalt binder sample is brought to the standard test temperature of 77°F (25°C) in a temperature-controlled water bath. The sample container is then placed in the penetrometer equipment. A needle of prescribed dimensions is attached to the penetrometer and suspended directly over the asphalt binder sample. A 50-gram weight is attached to the needle's loading platform so that the total weight used for loading is 100 grams (50-gram weight plus needle weight of 50 grams). The penetrometer is lowered until the needle tip just contacts the surface of the asphalt binder. The load is then released, allowing the weighted needle to penetrate the asphalt binder for 5 seconds. The distance that the needle penetrates into the asphalt binder is reported as the penetration value. This distance is reported in units of 0.1 millimeter, or deci-millimeter (dmm) as shown in Figure 2.7.



Figure 2.6 Penetrometer

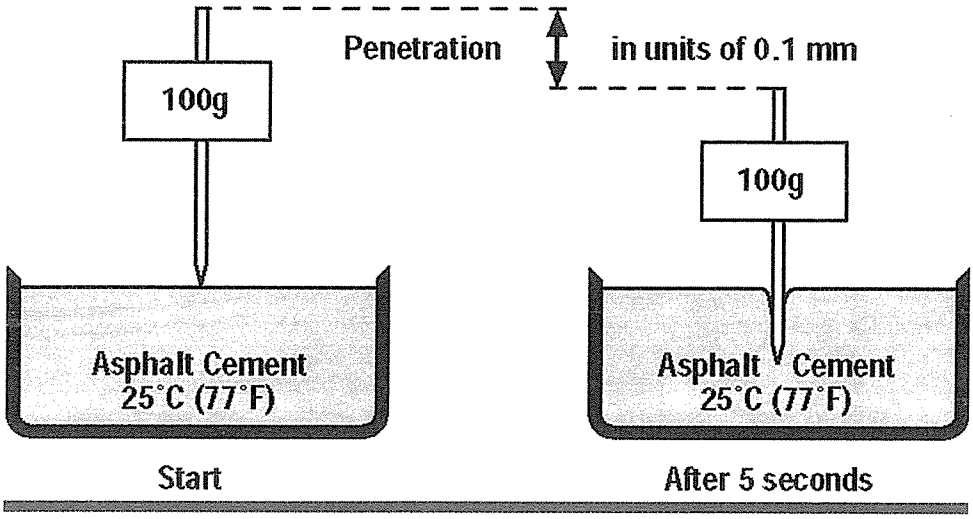


Figure 2.7 Penetration Test

A typical penetration test consists of three measurements made on the sample asphalt binder sample that are averaged to provide a single test value. Between each test, the needle is replaced with a clean needle and the sample container is rotated so that a different portion of the sample is tested.

Occasionally, the penetration test is conducted at a different test temperature, such as 41°F (4°C). In these instances, total load, penetration time, or both, may be varied.

Flash Point

As a petroleum product, an asphalt binder will release combustible fumes when heated to sufficiently high temperatures. The flash point provides an indication of the temperature at which a heated asphalt binder sample will instantaneously flash in the presence of an open flame. It is important to note that this temperature is usually well below the temperature at which the material will support combustion. The temperature at which the material supports combustion is called the fire point. It is rarely, if ever, used in asphalt binder specifications.

The most common test method used to determine the flash point of an asphalt binder is the Cleveland Open Cup (COC) flash point test (see Figure 2.8). In this test, a brass cup is filled with a specified volume of asphalt binder and is heated at a constant rate. A small flame is passed over the surface of the asphalt at timed intervals. When the flame passing over the sample surface causes an instantaneous flash, the temperature is recorded as the material's flash point.

Aging

Asphalt technologists have recognized that while the physical properties of some asphalt binders may be the same initially, the oxidation and volatilization that occurs during use in a hot mix facility and subsequent placement on the road might lead to a different in-service performance.

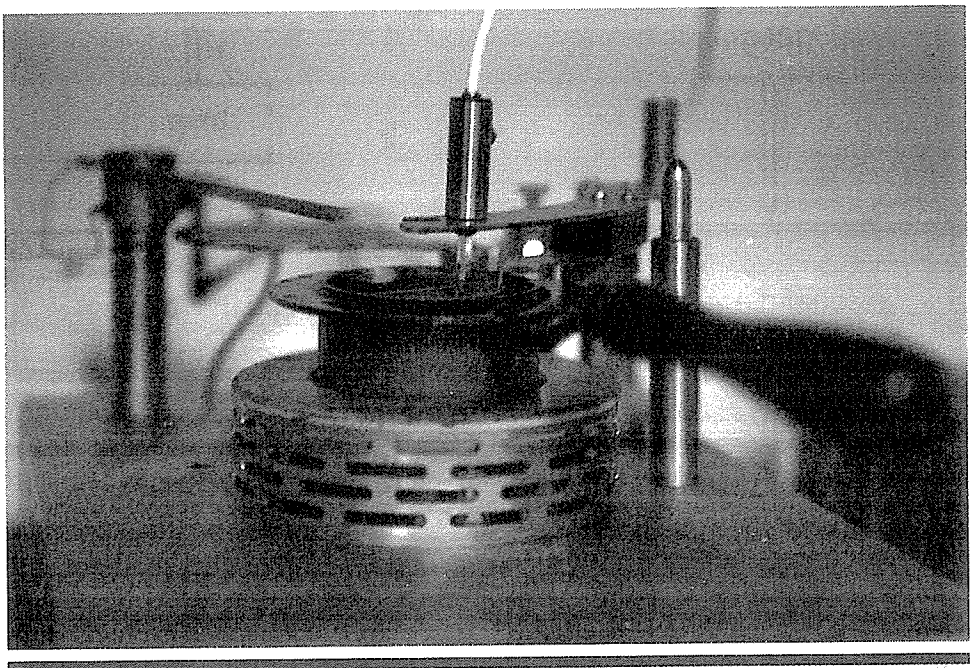


Figure 2.8 Cleveland Open Cup Test

Stated more simply, not all asphalt binders age the same and this difference in aging might result in a difference in actual performance.

Aging procedures were developed to subject an asphalt binder sample to hardening conditions that approximate the conditions that occur in normal, hot mix facility operations. The Thin Film Oven Test (TFOT) and Rolling Thin Film Oven Test (RTFOT) are two common short-term aging procedures. Historically, the TFOT had seen more widespread use than the RTFOT—which principally had been used in the western United States. The Performance Graded (PG) Asphalt Binder Specification now requires the use of the RTFOT. This will be discussed in more detail in the next section.

To address the issue of volatilization—the process of lighter volatile fractions being removed from the asphalt binder due to heating at elevated temperatures—both the TFOT and RTFOT procedures incorporate a mass change determination. In this determination, the test container (pan or bottle, depending on the procedure as described in the following text) is weighed initially. It is weighed again after the asphalt binder sample is added, and a third time after the completion of the test. The change in mass is calculated based on the weight of the sample before and after the aging procedure. It is expressed as a percentage of the mass before aging.

For most asphalt binders, the exposure to elevated temperatures and air results in volatilization and a loss of mass during the test. Usually, asphalt binders that are softer at room temperature might be expected to experience greater mass loss than stiffer asphalt binders. While uncommon, some asphalt binders experience a mass gain during the TFOT or RTFOT aging procedures. In these cases, the combination of heat and oxygen interacts in such a way with the asphalt binder that there is an increase in the mass.

The TFOT procedure is performed by pouring a 50-milliliter (approximately 50 grams) sample of heated asphalt binder into a flat-bottom, circular sample pan with an inside diameter of 5.5 inches (140 millimeters) and a depth of ⅜ inch (10 millimeters). After pouring, the layer of asphalt binder is approximately ⅛ inch (3 millimeters) thick. As noted earlier, if a mass change determination is needed, then the sample pans are weighed before and after the addition of the asphalt binder. The number of sample pans needed depends on the amount of sample needed for further testing after aging.

To start the test, the sample pans are placed on a rotating shelf in a ventilated oven (see Figure 2.9) operating at 325°F (163°C). The rotating shelf turns at a rate of approximately five to six revolutions per minute for a total testing time of five hours. After five hours, any sample pans being used for determining mass change are cooled to room temperature before the final weight is determined. The remaining pans are then poured and scraped into a single sample container for additional testing, such as viscosity or penetration.

The RTFOT procedure has the same purpose as the TFOT, but equipment and test procedures are different. Many asphalt technologists believe the

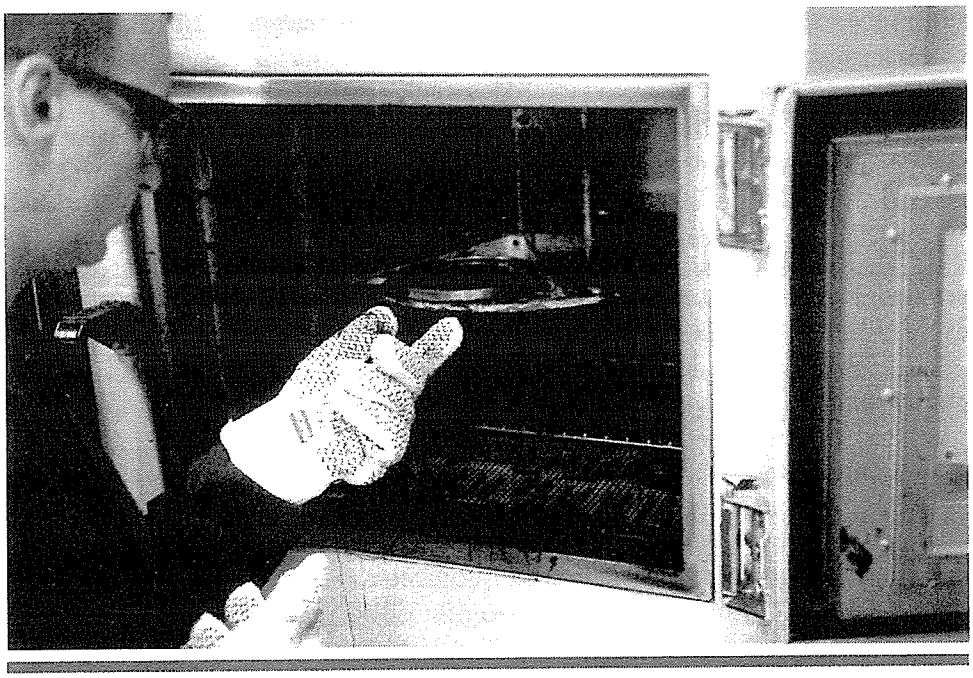


Figure 2.9 Thin Film Oven

RTFOT procedure provides slightly more aging (i.e., stiffness or viscosity) than the TFOT procedure.

The RTFOT procedure is performed by pouring 35 grams of heated asphalt binder into a glass bottle. As noted earlier, if a mass change determination is needed, then two designated RTFOT bottles are weighed before and after the addition of the asphalt binder. These bottles are then used to determine mass change. The number of bottles needed depends on the amount of sample needed for further testing after aging.

To start the test, the sample bottles are placed in a vertically rotating carriage in an oven (see Figure 2.10) operating at 325°F (163°C). The bottle carriage rotates at 15 revolutions per minute for a total testing time of 85 minutes. During the rotation, a jet of air flowing at a rate of 4000 milliliters per minute blows into each bottle as it passes the bottom position in the carriage. After 85 minutes, any sample bottles being used for determining mass change are cooled to room temperature before the final weight is determined (see Figure 2.11). The remaining bottles are then poured and scraped into a single sample container for additional testing.

Ductility

Ductility, as a physical property, has been considered an important characteristic of asphalt binders by some engineers. The presence or absence of ductility, however, is often considered more significant than the actual degree of ductility because some asphalt binders having a high degree of ductility have also been found to be more temperature-susceptible.

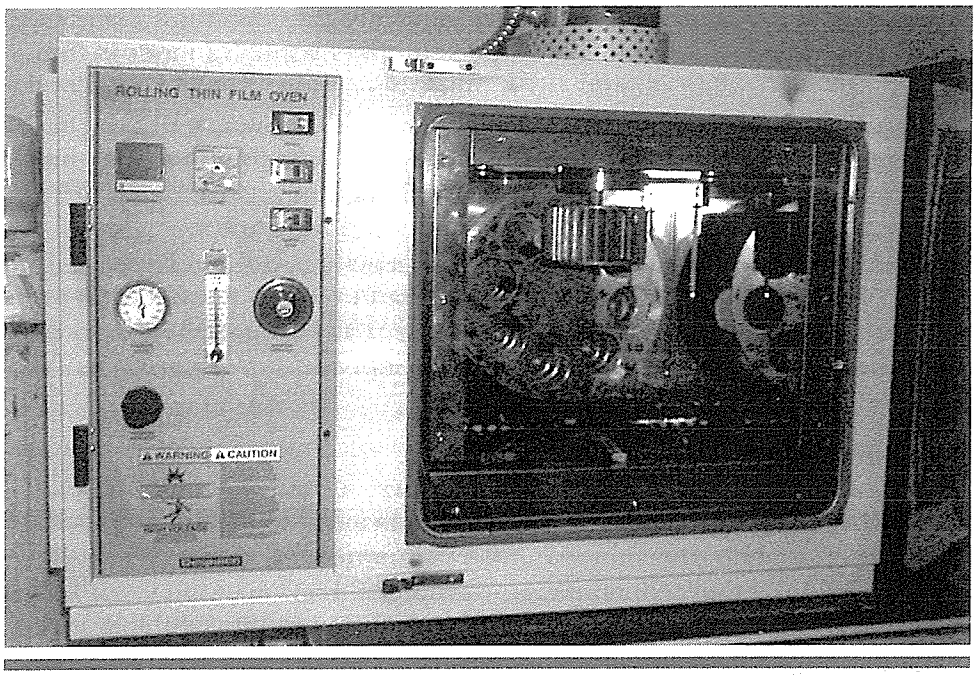


Figure 2.10 Rolling Thin Film Oven

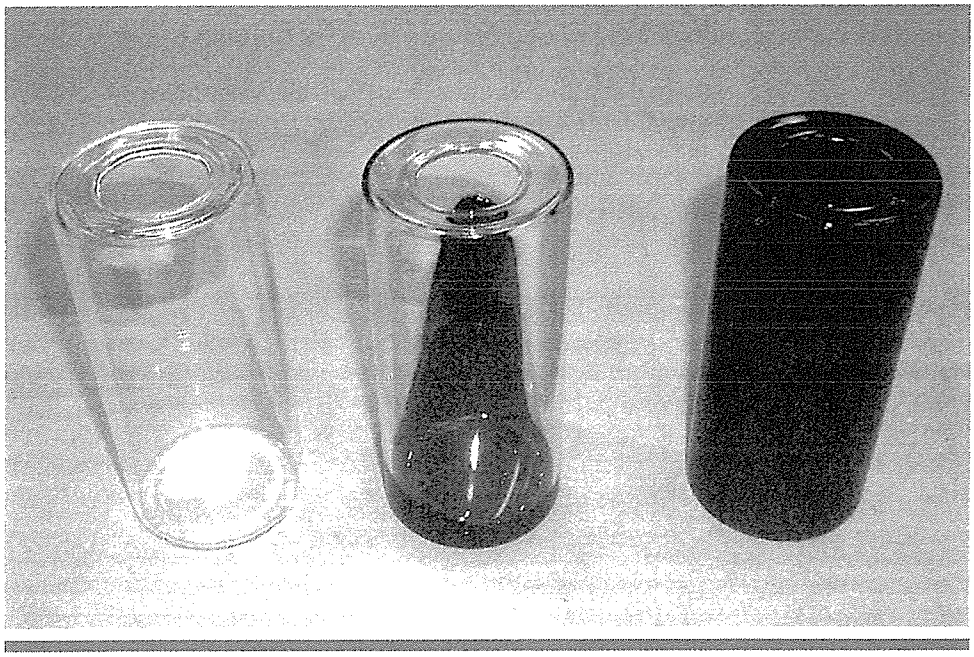


Figure 2.11 Rolling Thin Film Oven Bottles

The ductility test procedure is performed by first creating a standard mold shaped like a dog bone. A heated sample of asphalt binder is then poured into the mold, slightly overfilling the mold. After a prescribed cooling period, the specimen is trimmed flush with the surface of the mold (see Figure 2.12). The test specimen is then placed in the ductility water bath and conditioned to the desired test temperature, usually 77°F (25°C). The specimen is then loaded in the ductility machine and one end of the specimen is pulled away from the other at a specified rate of speed, normally 5 centimeters per minute until the thread of asphalt connecting the two parts of sample breaks (see Figure 2.13). The elongation (expressed in centimeters) at which the thread of material breaks is designated as the ductility of the asphalt.

Solubility

The solubility test is a measure of the purity of the asphalt binder. The portion of the asphalt binder that is soluble in carbon disulfide represents the active cementing constituents. Inert components—such as salts, free carbon, or non-organic contaminants—are insoluble. Due to the hazardous nature of carbon disulfide, trichloroethylene is used in the solubility test.

Determining the solubility of an asphalt binder involves dissolving approximately 2 grams of asphalt in 100 milliliters of solvent, and then filtering the solution through a filter paper placed in a porcelain (Gooch) crucible with holes in the bottom to allow fluid to pass. The amount of material retained on the filter is determined by weighing and is expressed as a percentage of the original sample weight.

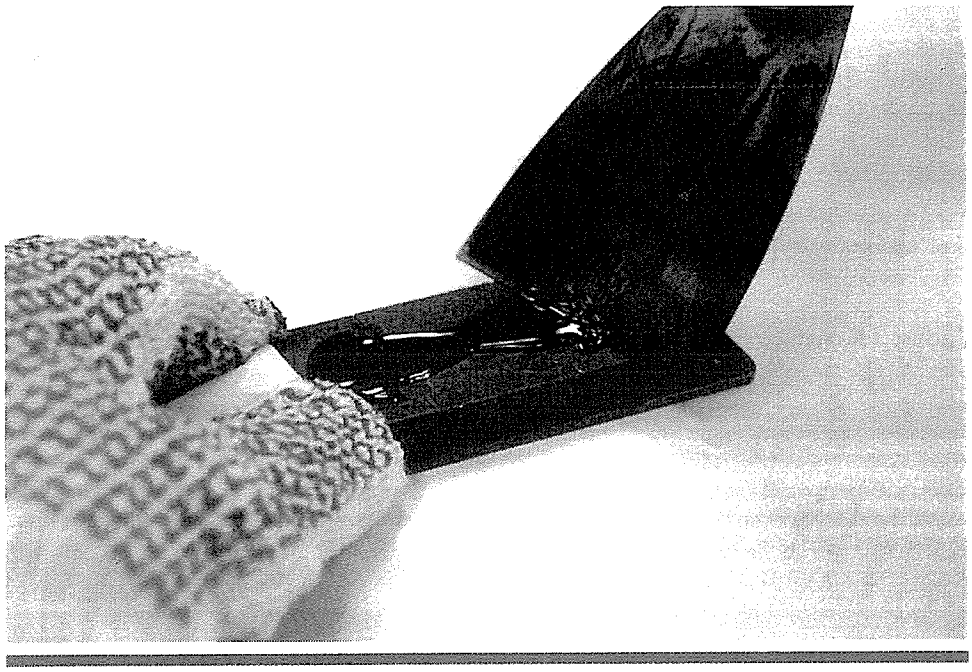


Figure 2.12 Trimming Ductility Specimen

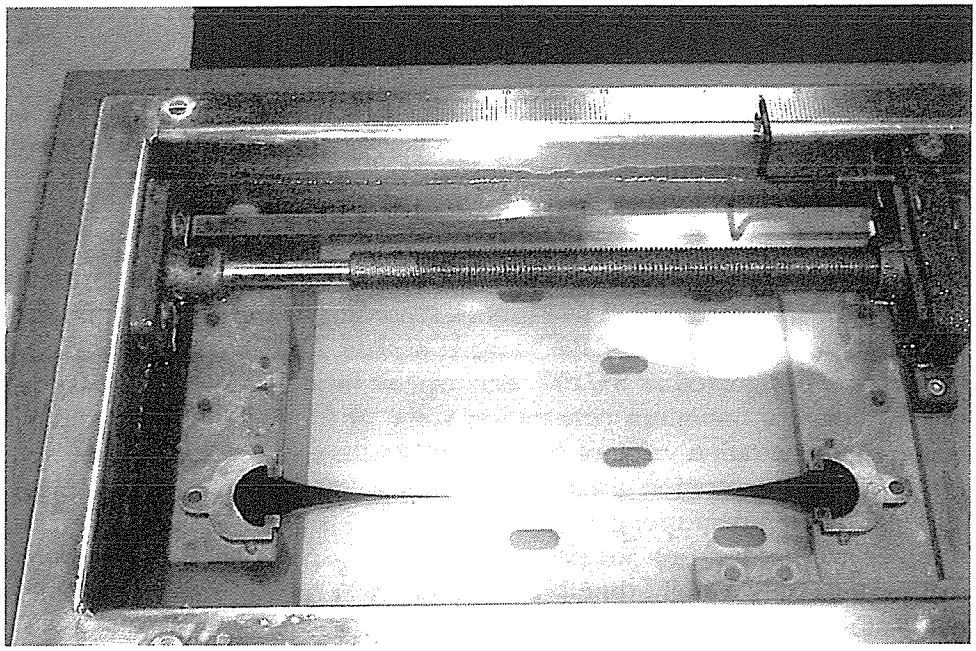


Figure 2.13 Ductility Test

Specific Gravity

The specific gravity of an asphalt binder can be used to convert between mass and volume. Knowing the asphalt binder specific gravity allows a user to calculate the mass of asphalt in a tank if the volume of asphalt in the tank is known. Specific gravity is also used in the calculation of some asphalt mixture properties.

The specific gravity test is performed by pouring heated asphalt binder into a pre-weighed, pre-calibrated glass pycnometer. After cooling, freshly boiled distilled or de-ionized water is added to the fill line. The pycnometer top is then lightly inserted and the pycnometer and sample are partially immersed in a water bath so that the sample acquires the appropriate test temperature. The pycnometer is removed from the water bath, wiped carefully, and weighed. Specific gravity is usually determined at either 60°F (15.6°C) or 77°F (25°C).

2.2.3 Specifications for Asphalt Cements

In 1903, A. W. Dow presented a paper describing the use of various tests, including solubility, in a specification for ensuring the quality of asphalts. At that time, most of the asphalts in use were natural, not petroleum, asphalt binders containing significant amounts of mineral matter. Thus, the amount of insoluble organic and inorganic matter in a given sample was a way to identify the source of the asphalt binder. As asphalts

from more than one source began to be used, specifications were written to attempt to control the quality of the asphalt binder.

Penetration-Graded

The ASTM D 946 specification was first established in 1947 to attempt to control the quality of the asphalt binder. Since the penetration value was the determining factor in identifying an asphalt binder's grade, the specification became known as the Penetration-Graded Specification.

In the Penetration Grading system, ASTM D 946 and AASHTO M 20, asphalt binders are classified into grades according to the penetration value measured at 77°F (25°C) of the unaged asphalt binder. Thus, an 85–100 (sometimes designated as 85/100) Pen asphalt binder would be an asphalt binder having a minimum penetration value of 85 deci-millimeters and a maximum penetration value of 100 deci-millimeters at 77°F (25°C). The softest (200–300) penetration grade asphalt is moderately firm at room temperature; at this temperature, gentle finger pressure indents the surface of the sample. The hardest (40–50) penetration grade asphalt is of a consistency that permits only a slight thumb print under firm pressure when the material is at room temperature.

Advantages and Shortcomings

While the Penetration-Graded Specification provided a means to evaluate the stiffness of an asphalt binder, it had some limitations. First, penetration is an empirical test that generally measures a combination of viscous and elastic behavior in an asphalt binder. It is an empirical test in which the results must be correlated with pavement performance in order to understand the value of the result. The relationship between the test result and performance may not be very good, depending on the quality and quantity of the data used in the relationship.

Second, the Penetration-Graded Specification only measured asphalt stiffness at an intermediate temperature. How stiff the asphalt binder might be in typical summer and winter temperatures could only be inferred from the intermediate temperature. As conceptually illustrated in Figure 2.14, the actual performance could be significantly different for asphalt binders with the same grade.

Finally, while short-term aging effects were captured through the use of the TFOT, there was no provision to conduct long-term aging of asphalt binders to simulate what happens to the asphalt stiffness after being in a pavement for a number of years.

Viscosity-Graded

Beginning in the 1970s, viscosity began to be used more frequently than penetration in the grading of asphalt binders. ASTM D 3381 and AASHTO M 226 established specification criteria using absolute viscosity at 140°F (60°C) as the principal physical property for grading. Viscosity at 275°F (135°C) is also usually specified. The purpose of the two criteria is to prescribe limiting values of consistency at two important temperatures. The 140°F (60°C) temperature was chosen as an approximation of the maximum temperature

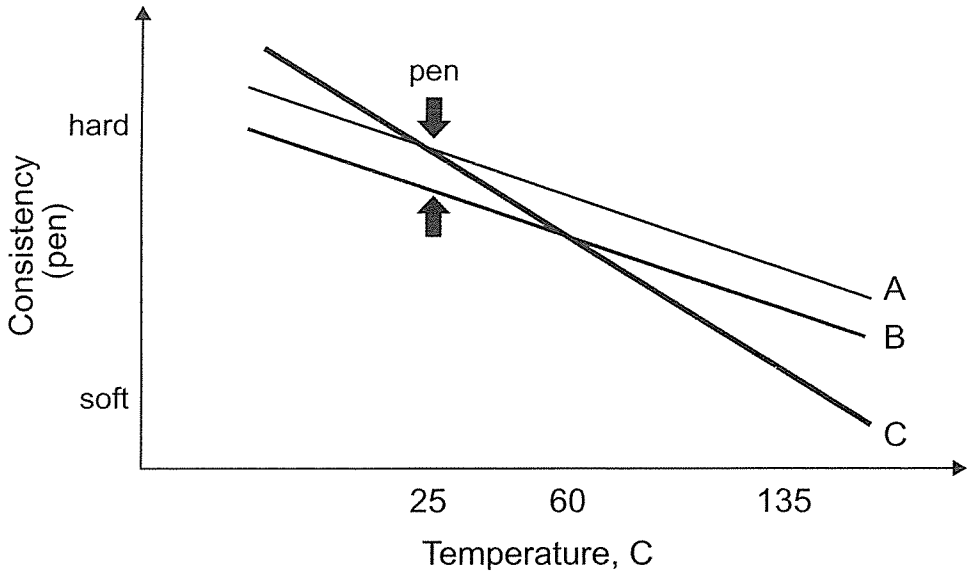


Figure 2.14 Asphalt Binder Properties—Penetration Grading

of asphalt pavement surfaces in service in the United States and elsewhere in the world. The 275°F (135°C) temperature was chosen as an approximation of the laydown temperature for HMA. Thus, the specification addresses asphalt binder consistency during the laydown process of HMA, when workability is an issue, and consistency at summer pavement temperatures, when stiffness is needed.

The AC System

In the Viscosity-Graded Specification, there are three tables that can be used to grade asphalt binders. Tables 1 and 2 use the AC (Asphalt Cement) system consisting of AC-2.5, AC-5, AC-10, AC-20, AC-40, and sometimes AC-30 grades. The numerical values in the grade designation denote the viscosity of the asphalt binder in hundreds of poises at 140°F (60°C). Thus, an AC-10 would be expected to have a viscosity of approximately 1000 poises at 140°F (60°C). In each grade, the allowable tolerance on viscosity is ± 20 percent. Thus an AC-10 grade would include asphalt binders with viscosity of 800–1200 poises at 140°F (60°C).

The AR System

Table 3 in the Viscosity-Graded Specification uses the AR (Aged Residue) system consisting of AR-1000, AR-2000, AR-4000, AR-8000, and AR-16000 grades. Like the AC system, the numerical values in the grade designation denote the viscosity in poises at 140°F (60°C), but with the viscosity being measured after the asphalt has been subjected to the rolling thin film oven test. In each grade, the allowable tolerance on viscosity is ± 25 percent. Thus an AR-4000 grade would include asphalt binders with viscosity, after the RTFOT, of 3000–5000 poises at 140°F (60°C).